TASK 4 – Write Correct Expressions

Expressions

fat_steps = round((fat - REFERENCE_FAT) / STEP_SIZE)
snf_steps = round((snf - REFERENCE_SNF) / STEP_SIZE)
rate = BASE_RATE + fat_steps * FAT_STEP_VALUE + snf_steps * SNF_STEP_VALUE
amount = round(litres * rate, 2)

The step count is signed: a quality reading below the reference produces a negative step count. Validate the values before calculating the rate.

Precedence and surprising results

Exponentiation ** has higher precedence than unary minus on its left: -2**2 is -(2**2), which is -4, while (-2)**2 is 4. Exponentiation is right-associative: 2**3**2 is 2**(3**2), or 512. Parentheses should be used to make intended grouping clear. Floating point can produce surprising values: (4.2 - 3.5) / 0.1 may be 7.000000000000001; 0.1 + 0.2 == 0.3 is False; and 0.3 - 0.2 may be 0.09999999999999998. round((4.2 - 3.5) / 0.1) gives the intended integer step count 7.

round, int and floor

round(x) rounds to the nearest integer, with ties handled using Python’s round-half-to-even rule. int(x) truncates toward zero. math.floor(x) rounds down toward negative infinity. For example, round(-1.6) is -2, int(-1.6) is -1 and math.floor(-1.6) is -2. For positive values such as 7.000000000000001, int and floor happen to give 7; however, for a negative step near -7 they can differ if floating-point error places the value slightly below the intended integer.

Comparison chaining and Boolean logic

The expression 2.0 <= fat <= 7.0 is equivalent to fat >= 2.0 and fat <= 7.0. Use and when both conditions must be true, or when either condition is sufficient, and not to reverse a condition. For example, litres > 0 and 2.0 <= fat <= 7.0 checks quantity and fat range.

Alternative calculation and complexity

For fat 4.2, the mathematical step value is +7; for fat 3.1 it is -4. With float arithmetic the raw values may be slightly above or below those integers. Applying round() gives +7 and -4. Applying int() or floor() to the positive raw value gives 7; for the negative raw value, truncation may give -3 if the representation is -3.999999999999996, while floor gives -4. That changes the rate by one fat step (Rs 0.40 per litre). The exact raw result can be checked in the Python version used. Each method performs a fixed number of arithmetic operations: O(1) time and O(1) extra space. Decimal arithmetic or scaled integer readings avoid relying on binary float for the step decision.